

Quadrilaterals, Perimeter & Area

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **calculate** interior or exterior angle measures of a polygon.

RELATED SKILLS

I Calculate polygon interior or exterior angle sums.

A Use inverse operations to maintain an equivalent equation.

- ☐ 2. I can **calculate** perimeter and area of common polygons.

RELATED SKILLS

I Add boundary lengths without omitting or duplicating sides.

D Select and report appropriate linear, square, or angular units.

A Apply the order of operations to a numerical expression.

- ☐ 3. I can **apply** properties of parallelograms to determine unknown measures.

RELATED SKILLS

I Apply parallelogram properties and tests.

A Translate quantities and relationships into an equation.

- ☐ 4. I can **determine** whether a quadrilateral is a parallelogram.

RELATED SKILLS

D Apply parallelogram properties and tests.

A Recognize and apply the converse of a conditional statement.

- ☐ 5. I can **apply** properties of rectangles, rhombi, and squares.

RELATED SKILLS

I Distinguish and apply rectangle, rhombus, and square properties.

I Use the inclusive hierarchy of quadrilateral classifications.

- ☐ 6. I can **classify** a quadrilateral from its stated or marked properties.

RELATED SKILLS

I Recall the definitions and defining properties of the major quadrilateral families.

D Use the inclusive hierarchy of quadrilateral classifications.

A Distinguish stated or marked facts from visual appearance.

☐ **7. I can **apply** trapezoid and trapezoid-midsegment relationships.**

RELATED SKILLS

- ☐ I Apply trapezoid and midsegment relationships.
 - ☐ A Use inverse operations to maintain an equivalent equation.
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☐ **8. I can **apply** properties of kites.**

RELATED SKILLS

- ☐ I Apply kite properties.
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☐ **9. I can **calculate** the perimeter or area of a composite polygon.**

RELATED SKILLS

- ☐ I Decompose a composite region into familiar shapes.
 - ☐ D Add boundary lengths without omitting or duplicating sides.
 - ☐ A Select and report appropriate linear, square, or angular units.
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☐ **10. I can **justify** a quadrilateral conclusion using definitions and theorems.**

RELATED SKILLS

- ☐ D Apply a geometric definition as justification.
 - ☐ D Select a theorem that connects known facts to a target statement.
 - ☐ A Organize statements and reasons into a valid proof sequence.
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